

3-Pump Core Equations

Three-pump rolling MILP and economic interpretation

Fixed-flow synthetic pumps, five-minute control intervals, and a 24-hour prediction horizon.

中文翻译：

三泵核心公式：三泵滚动 MILP 与经济性解释。三台固定流量合成测试泵、五分钟控制间隔，以及 24 小时预测时域。

Key updates from 1-pump to 3-pump

Component	1-pump	3-pump
Pump model	One fixed-flow pump	Three independent fixed-flow pumps
Decision variable	Scalar binary $u(k)$	Binary vector $\mathbf{u}(k) = [u_1, u_2, u_3]$
Optimization	Single-pump binary dynamic programming	Three-pump MILP solved by <code>intlinprog</code>
Available modes	Off or P1	All eight three-pump combinations
Switching	Change in one pump state	Separate transition variable for every pump
Controller output	One immediate request	Full $N \times 3$ plan plus first-step request
Execution feedback	One actual state and flow	Three actual states, mode, total flow, total power, and tracking error
Closed-loop memory	Previous scalar action	Previous <i>actually executed</i> 1×3 pump state

The outer rolling structure is unchanged: five-minute sampling, a 288-step horizon, re-optimization at every interval, and execution of only the first optimized action.

中文翻译：

从单泵到三泵的关键更新：

- 水泵模型：由一台固定流量泵升级为三台相互独立的固定流量泵。
- 决策变量：由标量二进制变量 $u(k)$ 扩展为 $\mathbf{u}(k) = [u_1, u_2, u_3]$ 。
- 优化方法：由单泵二进制动态规划升级为使用 `intlinprog` 求解的三泵 MILP。
- 运行模式：由“全关或 P1 开启”扩展为全部八种三泵组合。
- 启停处理：每台泵分别设置启停变化变量。
- 控制器输出：由单个即时请求扩展为完整的 $N \times 3$ 计划以及第一步请求。
- 执行反馈：返回三台泵的实际状态、模式、总流量、总功率和跟踪误差。
- 闭环记忆：下一轮使用上一时刻实际执行的 1×3 泵状态，而不是上一轮计划值。

外层滚动结构保持不变：每五分钟采样一次，预测时域为 288 步，每个时段重新优化，并且只执行最优序列的第一步。

Water-demand calculation

The hourly Richmond demand multipliers are converted into five-minute values using

$$d_{m,5\min} = \text{repelem}(d_{m,\text{hourly}}, 12) .$$

The water demand during interval k is

$$d(k) = d_{\text{base}} d_m(k) = 0.025 d_m(k) .$$

Where:

- $d(k)$: water demand during interval k , in m^3/s ;
- $d_{\text{base}} = 0.025 \text{ m}^3/\text{s}$: base demand, equivalent to 25 L/s;
- $d_m(k)$: dimensionless Richmond demand multiplier.

For example, if $d_m(k) = 1.61$, then $d(k) = 0.025 \times 1.61 = 0.04025 \text{ m}^3/\text{s} = 40.25 \text{ L/s}$.

中文翻译：

用水需求计算。 Richmond 每小时用水倍率通过 $d_{m,5\min} = \text{repelem}(d_{m,\text{hourly}}, 12)$ 转换成五分钟数据。第 k 个时段的需求为 $d(k) = d_{\text{base}} d_m(k) = 0.025 d_m(k)$ 。其中， $d(k)$ 是用水需求 (m^3/s)， $d_{\text{base}} = 0.025 \text{ m}^3/\text{s}$ 是基础需求，等于25 L/s， $d_m(k)$ 是无量纲 Richmond 需求倍率。例如，当 $d_m(k) = 1.61$ 时，需求为 $0.04025 \text{ m}^3/\text{s} = 40.25 \text{ L/s}$ 。

3-pump binary decisions

For pump $j \in \{1, 2, 3\}$ and prediction step k ,

$$u_j(k) \in \{0, 1\}, \quad 0 = \text{off}, \quad 1 = \text{on}.$$

The synthetic pump parameters used by 3-pump are

$$\mathbf{q} = [0.050 \quad 0.030 \quad 0.020] \text{ m}^3/\text{s}, \quad \mathbf{P} = [43 \quad 27 \quad 19] \text{ kW}.$$

Therefore, the total requested flow and power are

$$q(k) = \sum_{j=1}^3 q_j u_j(k), \quad P(k) = \sum_{j=1}^3 P_j u_j(k).$$

- The pumps have fixed flow and fixed power; there is no continuous pump-speed variable.
- All $2^3 = 8$ on/off combinations are available because $\sum_j u_j(k) \leq 3$.
- These values are synthetic test parameters, not measured Richmond pump data.

中文翻译：

三泵二进制决策。对水泵 $j \in \{1, 2, 3\}$ 和预测步 k , $u_j(k) \in \{0, 1\}$, 其中 0 表示关闭, 1 表示开启。三台泵的固定流量分别为 $[0.050, 0.030, 0.020] \text{ m}^3/\text{s}$, 固定功率分别为 $[43, 27, 19] \text{ kW}$ 。总请求流量为 $q(k) = \sum_{j=1}^3 q_j u_j(k)$, 总请求功率为 $P(k) = \sum_{j=1}^3 P_j u_j(k)$ 。本模型没有连续转速变量, 允许全部 $2^3 = 8$ 种开关组合。这些参数是合成测试参数, 不是 Richmond 实测泵参数。

All eight operating modes

Mode	Pump combination	$[u_1 \ u_2 \ u_3]$	Flow (L/s)	Power (kW)	Energy/5 min (kWh)	PE100
0	All off	$[0 \ 0 \ 0]$	0	0	0.000000	–
1	P1	$[1 \ 0 \ 0]$	50	43	3.583333	86.0000
2	P2	$[0 \ 1 \ 0]$	30	27	2.250000	90.0000
3	P3	$[0 \ 0 \ 1]$	20	19	1.583333	95.0000
4	P1+P2	$[1 \ 1 \ 0]$	80	70	5.833333	87.5000
5	P1+P3	$[1 \ 0 \ 1]$	70	62	5.166667	88.5714
6	P2+P3	$[0 \ 1 \ 1]$	50	46	3.833333	92.0000
7	P1+P2+P3	$[1 \ 1 \ 1]$	100	89	7.416667	89.0000

The five-minute electrical energy is

$$E_{\text{step}} = P \frac{300}{3600} = \frac{P}{12} \text{ in kWh.}$$

PE100 is an efficiency-display metric, $PE100 = 100P/q_{L/s}$ in $\text{kW}/(100 \text{ L/s})$. It is **not** used to calculate the electricity bill.

中文翻译：

全部八种运行模式。表格列出了全部八种运行模式：模式 0 为全关；模式 1、2、3 分别为 P1、P2、P3；模式 4、5、6 分别为 P1+P2、P1+P3、P2+P3；模式 7 为三泵全开。每个五分钟时段的耗电量为 $E_{\text{step}} = P(300/3600) = P/12$ kWh。PE100 按照 $100P/q_{L/s}$ 计算，单位为 $\text{kW}/(100 \text{ L/s})$ ，只用于效率展示，**不能**用于计算实际电费。

Tank-level dynamics

The tank level is updated using the actual net inflow:

$$x(k+1) = x(k) + \frac{\Delta t}{A} [q(k) - d(k)].$$

Substituting $\Delta t = 300$ s, $A = 1000$ m², and the three-pump flow equation gives

$$x(k+1) = x(k) + 0.3 [0.050u_1(k) + 0.030u_2(k) + 0.020u_3(k) - d(k)].$$

Where:

- $x(k)$: tank level at the beginning of interval k , in m;
- $x(k+1)$: level after one five-minute interval, in m;
- $A = 1000$ m²: tank cross-sectional area;
- $q(k)$: combined pump inflow, in m³/s;
- $d(k)$: water demand, in m³/s.

The level is a state determined by mass balance; it is not an independent control input.

中文翻译：

水箱水位动态。水箱水位按照实际净流入更新： $x(k+1) = x(k) + (\Delta t/A)[q(k) - d(k)]$ 。代入 $\Delta t = 300$ 秒、 $A = 1000$ m² 以及三泵流量后，得到 $x(k+1) = x(k) + 0.3[0.050u_1(k) + 0.030u_2(k) + 0.020u_3(k) - d(k)]$ 。 $x(k)$ 是时段开始时的水位， $x(k+1)$ 是五分钟后的水位， A 是水箱横截面积， $q(k)$ 是三泵合计流入， $d(k)$ 是用水需求。水位由质量平衡决定，不是可以独立选择的控制输入。

MILP constraints

At every step of the prediction horizon, the controller enforces

$$u_j(k) \in \{0, 1\}, \quad 1.40 \leq x(k) \leq 3.37, \quad \sum_{j=1}^3 u_j(k) \leq 3.$$

The tank dynamics must also hold exactly:

$$x(k+1) - x(k) = \frac{300}{1000} \left[\sum_{j=1}^3 q_j u_j(k) - d(k) \right].$$

- Initial level: $x(0) = 2.30$ m.
- Prediction horizon: $N = 288$ five-minute intervals, or 24 hours.
- No terminal-level equality is imposed.
- No flow-tracking penalty is added.
- Historical prices are used as perfect-forecast inputs to test the controller, not to validate a price predictor.

中文翻译：

MILP 约束。在预测时域的每一步，控制器都必须满足：泵状态为二进制变量，水位位于 1.40–3.37m 之间，同时运行的泵不超过三台，并且水箱质量平衡严格成立。初始水位为 2.30 m，预测时域为 $N = 288$ 个五分钟时段，即 24 小时。模型不设置终点水位等式，也不添加流量跟踪惩罚。历史电价作为完美预测输入，只用于验证控制器，不代表已经验证价格预测器。

Per-pump switching transitions

Each pump has a switching variable $s_j(k)$ that records whether its state changes. For the first predicted interval,

$$s_j(1) = |u_j(1) - u_{j,\text{previous}}|.$$

For later intervals,

$$s_j(k) = |u_j(k) - u_j(k-1)|, \quad k = 2, \dots, N.$$

Thus, $s_j(k) = 0$ when pump j keeps the same state and $s_j(k) = 1$ when it turns on or off.

- Switching cost: $c_s = 0.20$ AUD per transition per pump.
- An on transition and an off transition are counted separately.
- Changing from P1+P2 to P1+P3 produces two transitions: P2 turns off and P3 turns on.
- The previous state comes from the pumps actually executed in the preceding closed-loop interval.
- In the MATLAB MILP, the absolute value is implemented by two linear inequalities; the positive switching cost makes the optimum equal to the absolute difference shown above.

中文翻译：

逐泵启停变化。 每台泵都有一个启停变量 $s_j(k)$ ，用于记录泵状态是否发生变化。第一个预测步为 $s_j(1) = |u_j(1) - u_{j,\text{previous}}|$ ；后续预测步为 $s_j(k) = |u_j(k) - u_j(k-1)|$ 。如果泵状态不变，则 $s_j(k) = 0$ ；如果泵开启或关闭，则 $s_j(k) = 1$ 。每台泵每次 transition 的成本为 0.20 AUD。例如从 P1+P2 切换到 P1+P3 时，P2 关闭、P3 开启，因此共计两次 transition。下一轮优化的 previous state 来自上一时刻实际执行结果。MATLAB 的 MILP 代码仍使用两个线性不等式实现这个绝对值；正的启停成本会使最优解等于上面展示的绝对差。

Electricity-cost conversion

The electricity price $p(k)$ is in AUD h/MW, whereas pump power $P(k)$ is in kW. The five-minute energy cost is

$$C_{\text{energy}}(k) = p(k)P(k)\frac{300}{3600}\frac{1}{1000}.$$

Unit check:

$$\frac{\text{AUD}}{\text{MWh}} \times \text{kW} \times \text{h} \times \frac{\text{MW}}{1000 \text{ kW}} = \text{AUD}.$$

For example, one five-minute P1 interval uses

$$E_{1,\text{step}} = 43 \times \frac{300}{3600} = 3.583\,333 \text{ kW h} = 0.003\,583\,333 \text{ MW h}.$$

The factor 1/1000 must be retained. PE100 is used only for efficiency display; the electricity bill uses the actual powers 43, 27, and 19 kW.

中文翻译：

电费单位换算。电价 $p(k)$ 的单位为AUD h/MW，泵功率 $P(k)$ 的单位为 kW。五分钟电费为 $C_{\text{energy}}(k) = p(k)P(k)(300/3600)(1/1000)$ 。300/3600 把五分钟换算为小时，1/1000 把 kW 换算为 MW，这个因子不能省略。例如 P1 运行五分钟消耗3.583 333 kW h，即 0.003 583 333 MW h。实际电费使用 43、27 和 19 kW，PE100 只用于展示效率，不参与电费计算。

Three-pump MPC objective function

At every sampling instant, the MILP minimizes predicted electricity and per-pump switching costs over $N = 288$ future intervals:

$$\min_{u,s} J = \sum_{k=1}^N \left[p(k) \left(\sum_{j=1}^3 P_j u_j(k) \right) \frac{300}{3600} \frac{1}{1000} + c_s \sum_{j=1}^3 s_j(k) \right].$$

Where:

- J : predicted operating cost over the 24-hour horizon, in AUD;
- $p(k)$: predicted electricity price, in AUD h/MW;
- P_j : fixed power of pump j , in kW;
- $u_j(k)$: planned binary state of pump j ;
- $s_j(k)$: switching transition of pump j ;
- $c_s = 0.20$ AUD per transition per pump.

There is no separate penalty on pumped volume or final tank level.

中文翻译：

三泵 MPC 目标函数。在每个采样时刻，MILP 在未来 $N = 288$ 个时段内最小化预测电费与逐泵启停成本。 J 表示 24 小时预测时域总运行成本； $p(k)$ 是预测电价； P_j 是第 j 台泵的固定功率； $u_j(k)$ 是计划开关状态； $s_j(k)$ 是第 j 台泵的启停变化； $c_s = 0.20$ AUD 表示每台泵每次 transition 的成本。当前目标函数没有单独惩罚总泵水量，也没有惩罚或固定终点水位。

Rolling closed-loop MPC operation

At closed-loop interval t , 3-Pump performs the following sequence:

1. read the current measured tank level $x(t)$ and the previously executed pump state $\mathbf{u}_{\text{actual}}(t-1)$;
2. build a 288-step price and demand prediction window;
3. solve the three-pump MILP using `intlinprog`;
4. obtain the planned matrix $\mathbf{U}^* \in \{0, 1\}^{288 \times 3}$;
5. execute only the first row, $\mathbf{u}_{\text{request}}(t) = \mathbf{U}^*(1, :)$;
6. obtain actual pump states, total flow, and total power from the execution interface;
7. update the tank with actual flow and calculate cost using actual power;
8. advance five minutes and solve a new 288-step MILP.

Thus a 24-hour closed-loop test solves 288 separate MILPs; it is not one open-loop 24-hour schedule.

中文翻译：

滚动闭环 MPC 运行过程。在闭环时段 t ，3-Pump 首先读取当前实测水位 $x(t)$ 和上一时刻实际执行的泵状态，然后建立 288 步价格与需求预测窗口，使用 `intlinprog` 求解三泵 MILP，得到 $\mathbf{U}^* \in \{0, 1\}^{288 \times 3}$ 。控制器只执行第一行 $\mathbf{u}_{\text{request}}(t) = \mathbf{U}^*(1, :)$ ，随后从执行接口获得实际泵状态、总流量和总功率，用实际流量更新水位、用实际功率计算电费。五分钟后窗口向前滚动，再次求解新的 288 步 MILP。因此完整 24 小时闭环需要求解 288 次 MILP，并不是一次求出并执行完整 24 小时开环计划。

Why the solver does not enumerate 8^{288} schedules

At one future interval, three binary pumps produce eight possible operating modes. Therefore, a 288-step plan has

$$8^{288} = 2^{864}$$

theoretical mode sequences. This number describes the size of the search space; it is **not** the number of schedules explicitly tested by the MATLAB code.

Each 288-step MILP contains

- $3 \times 288 = 864$ binary pump-state variables $u_j(k)$;
- $3 \times 288 = 864$ continuous switching helper variables $s_j(k)$;
- linear constraints for tank level, mass balance, pump availability, and per-pump switching.

The optimizer considers all 288 future intervals together. It does not run a simple loop that tries eight modes at step 1, then eight modes at step 2, and so on until every complete path has been listed.

中文翻译：

为什么求解器不会枚举 8^{288} 条路径。 在一个未来时段内，三台二进制水泵共有八种运行模式，因此一个 288 步计划在理论上存在 $8^{288} = 2^{864}$ 种模式序列。这个数字表示搜索空间有多大，**不是** MATLAB 代码实际逐条测试的方案数量。每次 288 步 MILP 包含 864 个二进制泵状态变量 $u_j(k)$ 、864 个连续启停辅助变量 $s_j(k)$ ，以及水位、质量平衡、泵可用性和逐泵启停约束。优化器会把未来 288 步作为一个整体处理，而不是用普通循环把每一条完整路径全部列出来。

How intlinprog solves the 288-step MILP

The solution logic can be understood as “estimate first, then remove unpromising branches”.

1. **Temporarily allow partial pump states.** The binary condition is relaxed to $0 \leq u_j(k) \leq 1$ and an easier linear program is solved. A value such as $u_1(k) = 0.4$ is only a temporary estimate used to calculate the lowest possible cost.
2. **Split a fractional decision into two branches.** If $u_1(k) = 0.4$, the solver creates one branch with $u_1(k) = 0$ and another with $u_1(k) = 1$.
3. **Remove branches that cannot be cheaper.** If the best possible cost of a branch is already higher than the best feasible integer plan found so far, that branch is discarded.
4. **Remove branches that violate physical constraints.** A branch is also discarded if it must make the level fall below 1.40 m, rise above 3.37 m, or violate mass balance and pump constraints.

This is similar to route planning: estimate which directions may still produce a good route, then stop exploring directions that are impossible or already too expensive. The 24-hour closed-loop test repeats this 288-step MILP solve 288 times and executes only the first action after each solve.

中文翻译：

intlinprog 如何求解 288 步 MILP。它的逻辑可以理解为“先估计，再排除”：

1. 暂时允许水泵部分开启。先把二进制条件放宽为 $0 \leq u_j(k) \leq 1$ 并求解较简单的线性规划。像 $u_1(k) = 0.4$ 这样的数值只是临时估计，用于计算理论最低成本，不会被实际执行。
2. 把小数决策分成两个分支。如果出现 $u_1(k) = 0.4$ ，就分别检查 $u_1(k) = 0$ 和 $u_1(k) = 1$ 两个方向。
3. 删除不可能更便宜的分支。如果某个分支即使在最理想情况下也比当前最佳整数方案更贵，就直接停止搜索该分支。
4. 删除违反物理约束的分支。如果某个分支必然导致水位低于 1.40 m、高于 3.37 m，或者违反质量平衡和泵约束，就立即删除。

这类似导航软件：先估计哪些方向仍可能形成好路线，再放弃已经不可行或明显太贵的方向。完整 24 小时闭环会重复求解这个 288 步 MILP 共 288 次，每次只执行第一步。

Absolute cost versus cost per cubic metre

The historical comparison produced the following realised results:

Metric	Multi-pump	P1-only
Pumped volume	1830 m ³	1590 m ³
Final tank level	1.9781 m	1.7381 m
Absolute total cost	29.7296 AUD	27.7452 AUD
Cost per pumped volume	0.016 25 AUD/m ³	0.017 45 AUD/m ³

- **Absolute cost:** P1-only is cheaper because it pumps 240 m³ less water during the displayed day.
- **Cost per cubic metre:** P1-only is more expensive per unit of pumped water. Multi-pump is approximately 6.9% cheaper on this metric.
- The final inventories are different, so absolute cost alone is not an equal-service economic comparison.

中文翻译：

绝对成本与单位水量成本。历史闭环结果显示：多泵抽水1830 m³、终点水位 1.9781 m、绝对总成本29.7296 AUD；P1-only 抽水 1590 m³、终点水位1.7381 m、绝对总成本 27.7452 AUD。因此 P1-only 的绝对总成本更低，但它也少抽了 240 m³。按泵水量归一化后，多泵为 0.016 25 AUD/m³，P1-only 为 0.017 45 AUD/m³，所以多泵的单位水量成本约低 6.9%。两者终点库存不同，因此只比较绝对总成本并不是等服务量比较。

Why the two conclusions are both possible

The volume difference is exactly explained by final tank inventory:

$$\Delta V = A\Delta x = 1000(1.9781 - 1.7381) = 240 \text{ m}^3.$$

This equals the pumping difference $1830 - 1590 = 240 \text{ m}^3$: the extra water remained in the tank and was not extra demand.

- P1-only has the lower *total daily bill*.
- Multi-pump has the lower *cost per pumped cubic metre*, because more pumping is shifted into cheaper price intervals.
- The lower relative quantity is **AUD/m³**, not **kWh/m³**; therefore “relative consumption” is ambiguous.

The electrical intensities are

$$\text{Multi} = 0.2448 \text{ kWh/m}^3, \quad \text{P1-only} = 0.2389 \text{ kWh/m}^3.$$

Conclusion: the economic advantage per cubic metre comes from electricity-price timing, not from lower physical energy use.

中文翻译：

为什么两个结论可以同时成立。水量差完全由终点库存解释： $\Delta V = A\Delta x = 1000(1.9781 - 1.7381) = 240 \text{ m}^3$ ，正好等于 $1830 - 1590 = 240 \text{ m}^3$ 。多出的水储存在水箱中，不是额外需求。P1-only 的每日总账单更低，而多泵的每立方米成本更低，原因是它把更多抽水转移到了低电价时段。这里较低的相对指标是 **AUD/m³**，不是 **kWh/m³**。实际上多泵与 P1-only 的电能强度分别约为 0.2448 和 0.2389 kWh/m³。因此单位水量经济优势来自电价时序，而不是更低的物理单位能耗。

Interpretation and fair comparison

The multi-pump feasible set contains the P1-only actions, but this does not guarantee a lower truncated 24-hour realised bill when the two trajectories finish with different stored water.

- Every rolling optimization looks 24 hours ahead, including times beyond the end of the displayed day.
- With no terminal-level equality, the controller may use a cheap period today to store water for future demand.
- The reported comparison is fair in price, demand, initial level, constraints, horizon, and simulation duration, but not in final inventory.

For a strict economic comparison, report all three quantities together:

total cost, cost per pumped m³, final stored volume.

Alternatively, run a separate analysis in which both controllers are required to finish at the same tank level. This is an evaluation choice; it is not part of the current 3-Pump controller specification.

中文翻译：

结果解释与公平比较。多泵控制器的可行域包含 P1-only 动作，但当两条闭环轨迹具有不同终点库存时，这并不保证截取出来的 24 小时实际账单一定更低。每次滚动优化都会向未来看 24 小时，包括显示日结束之后的时段；由于没有终点水位等式，控制器可能利用今天的低价时段为未来需求储水。当前对照在电价、需求、初始水位、约束、预测时域和仿真长度方面一致，但终点库存不一致。严格比较应同时报告总成本、每立方米成本和最终储水量；也可以另做两种控制器终点水位相同的分析，但这属于评价方法，不是当前 3-Pump 控制器规范的一部分。